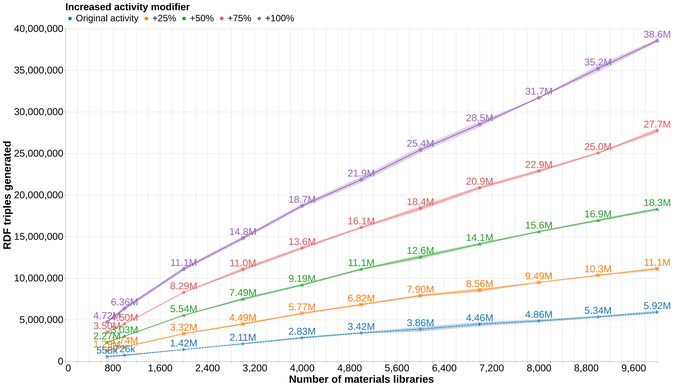
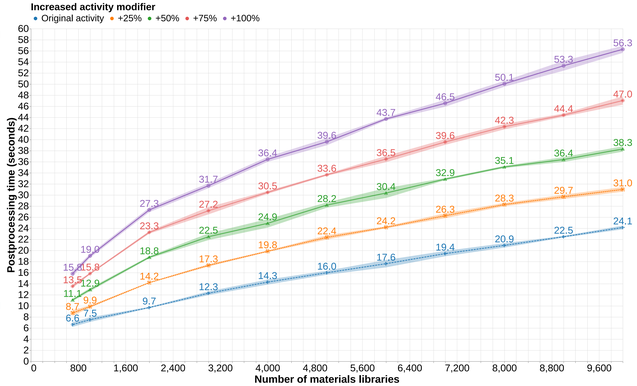
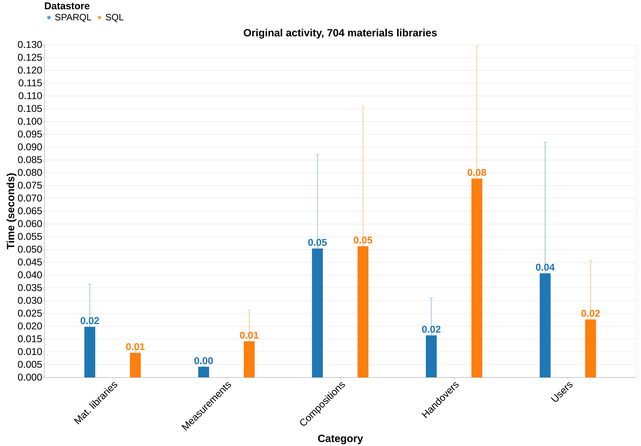
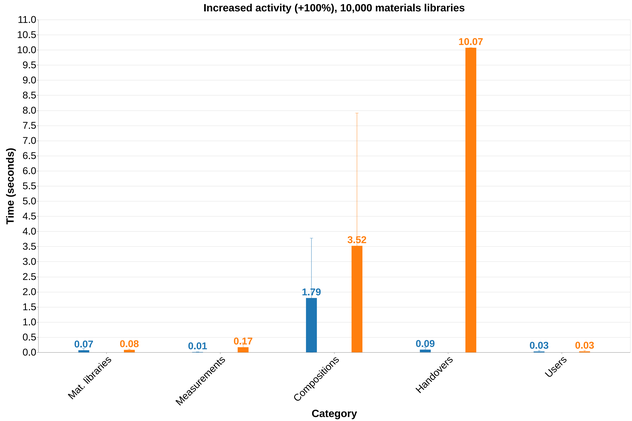

Increased activity modifier
Original activity
+25%
+50%
+75%
+100%
RDF triples generated
Number of materials libraries
0
5,000,000
10,000,000
15,000,000
20,000,000
25,000,000
30,000,000
35,000,000
40,000,000
0
800
1,600
2,400
3,200
4,000
4,800
5,600
6,400
7,200
8,000
8,800
9,600
6.36M
8.29M
11.1M
14.8M
18.7M
21.9M
25.4M
28.5M
31.7M
35.2M
38.6M
4.72M
6.50M
8.93M
11.0M
13.6M
16.1M
18.4M
20.9M
22.9M
25.0M
27.7M
3.50M
4.93M
7.49M
9.19M
11.1M
12.6M
14.1M
15.6M
16.9M
18.3M
2.27M
3.32M
5.54M
5.77M
6.82M
7.90M
8.56M
9.49M
10.3M
11.1M
1.28M
1.74M
4.49M
2.83M
3.42M
3.86M
4.46M
4.86M
5.34M
5.92M
556k
26k
1.42M
2.11M


Increased activity modifier
Original activity
+25%
+50%
+75%
+100%
Postprocessing time (seconds)
Number of materials libraries
0
2
4
6
8
10
12
14
16
18
20
22
24
26
28
30
32
34
36
38
40
42
44
46
48
50
52
54
56
58
60
0
800
1,600
2,400
3,200
4,000
4,800
5,600
6,400
7,200
8,000
8,800
9,600
19.0
27.3
31.7
36.4
39.6
43.7
46.5
50.1
53.3
56.3
15.8
23.3
27.2
30.5
33.6
36.5
39.6
42.3
44.4
47.0
13.5
18.8
22.5
24.9
28.2
30.4
32.9
35.1
36.4
38.3
11.1
14.2
17.3
19.8
22.4
24.2
26.3
28.3
29.7
31.0
8.7
9.7
12.3
14.3
16.0
17.6
19.4
20.9
22.5
24.1
6.6
7.5


Datastore
SPARQL
SQL
Original activity, 704 materials libraries
Time (seconds)
Category
0.000
0.005
0.010
0.015
0.020
0.025
0.030
0.035
0.040
0.045
0.050
0.055
0.060
0.065
0.070
0.075
0.080
0.085
0.090
0.095
0.100
0.105
0.110
0.115
0.120
0.125
0.130
0.02
0.01
0.00
0.01
0.05
0.05
0.02
0.08
0.04
0.02
Mat. libraries
Measurements
Compositions
Handovers
Users


Increased activity (+100%), 10,000 materials libraries
Time (seconds)
Category
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
5.5
6.0
6.5
7.0
7.5
8.0
8.5
9.0
9.5
10.0
10.5
11.0
0.07
0.08
0.01
0.17
1.79
3.52
0.09
10.07
0.03
0.03
Mat. libraries
Measurements
Compositions
Handovers
Users